Rockchip Android14 音频功能用户指南

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前言

概述

本文将指导用户如何配置和开发一些常见的音频扩展功能。

产品版本

Android 14

读者对象

本文档（本指南）主要适用于以下工程师：

技术支持工程师

软件开发工程师

修订记录

版本号	作者	修改日期	修改说明
V1.0.0	刘兴亮	2024-01-18	初始版本

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1. 设备类型扩展

1.1 需求分析

在system/media/audio/include/system/audio-hal-enums.h头文件中audio_devices_t中列出了所有谷歌支持的设备类型，比如：

speaker：AUDIO_DEVICE_OUT_SPEAKER

hdmi：AUDIO_DEVICE_OUT_HDMI

spdif：AUDIO_DEVICE_OUT_SPDIF

对于RK3588等平台，存在多个同类型的设备，且都对应同一个设备类型，比如hdmi0/hdmi1都对应设备类型AUDIO_DEVICE_OUT_HDMI，如果hdmi0和hdmi1同时插入，后被系统识别到的设备会将先被系统识别到的设备溢出，无法共存。另外，扩展类型不能影响GMS/ELDA认证。

1.2 google官方说明

https://source.android.google.cn/docs/core/audio/device-type-limit?hl=zh-cn

https://source.android.google.cn/docs/core/audio/hidl-implement?authuser=1&hl=zh-cn

1.3 扩展方法

hardware/interfaces/audio

```
commit 85bced3b4b0a59be8ade74151cde975b1132acab
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Thu Jan 5 16:51:37 2023 +0800

    Audio: add define for VX_ROCKCHIP_XXX

    Signed-off-by: shine.liu <shine.liu@rock-chips.com>
    Change-Id: I67ad316b687ef0b7cfd8638fe4c546db3e6a0330

diff --git a/audio/7.1/config/api/current.txt b/audio/7.1/config/api/current.txt
index 75fc5c044..d5dcb1499 100644
--- a/audio/7.1/config/api/current.txt
+++ b/audio/7.1/config/api/current.txt
@@ -569,6 +569,13 @@ package android.audio.policy.configuration.V7_1 {
     method public void setFormats(@Nullable
android.audio.policy.configuration.V7_1.SurroundFormats);
   }

+  public enum VendorExtension {
+    method @NonNull public String getRawName();
```

```
+    enum_constant public static final
android.audio.policy.configuration.V7_1.VendorExtension VX_ROCKCHIP_IN_HDMI0;
+    enum_constant public static final
android.audio.policy.configuration.V7_1.VendorExtension VX_ROCKCHIP_OUT_HDMI0;
+    enum_constant public static final
android.audio.policy.configuration.V7_1.VendorExtension VX_ROCKCHIP_OUT_SPDIF0;
+  }
+
   public enum Version {
     method @NonNull public String getRawName();
     enum_constant public static final
android.audio.policy.configuration.V7_1.Version _7_0;
diff --git a/audio/7.1/config/audio_policy_configuration.xsd
b/audio/7.1/config/audio_policy_configuration.xsd
index 7e1da9051..f71b7e71d 100644
--- a/audio/7.1/config/audio_policy_configuration.xsd
+++ b/audio/7.1/config/audio_policy_configuration.xsd
@@ -330,6 +330,9 @@
         -->
         <xs:restriction base="xs:string">
             <xs:pattern value="VX_[A-Z0-9]{3,}_[_A-Z0-9]+"/>
+            <xs:enumeration value="VX_ROCKCHIP_OUT_HDMI0"/>
+            <xs:enumeration value="VX_ROCKCHIP_OUT_SPDIF0"/>
+            <xs:enumeration value="VX_ROCKCHIP_IN_HDMI0"/>
         </xs:restriction>
     </xs:simpleType>
     <xs:simpleType name="extendableAudioDevice">
```

system/media

```
commit b6a3c79df123313dcacdabd90d94831990047c7d
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Thu Jan 5 16:46:55 2023 +0800

    Audio: add define for VX_ROCKCHIP_XXX

    Signed-off-by: shine.liu <shine.liu@rock-chips.com>
    Change-Id: Ibc908c2d66fd8258c9511d48981ff75d0590290d

diff --git a/audio/include/system/audio-base-utils.h
b/audio/include/system/audio-base-utils.h
index 4b815628..d19bc3e6 100644
--- a/audio/include/system/audio-base-utils.h
+++ b/audio/include/system/audio-base-utils.h
@@ -164,6 +164,7 @@ static CONST_ARRAY audio_devices_t
AUDIO_DEVICE_OUT_ALL_ARRAY[] = {
     AUDIO_DEVICE_OUT_BLUETOOTH_A2DP_HEADPHONES, // 0x00000100u
     AUDIO_DEVICE_OUT_BLUETOOTH_A2DP_SPEAKER,    // 0x00000200u
     AUDIO_DEVICE_OUT_HDMI,                      // 0x00000400u, OUT_AUX_DIGITAL
+    VX_ROCKCHIP_OUT_HDMI0,
     AUDIO_DEVICE_OUT_ANLG_DOCK_HEADSET,         // 0x00000800u
     AUDIO_DEVICE_OUT_DGTL_DOCK_HEADSET,         // 0x00001000u
     AUDIO_DEVICE_OUT_USB_ACCESSORY,             // 0x00002000u
@@ -174,6 +175,7 @@ static CONST_ARRAY audio_devices_t
AUDIO_DEVICE_OUT_ALL_ARRAY[] = {
```

```
     AUDIO_DEVICE_OUT_HDMI_ARC,                  // 0x00040000u
     AUDIO_DEVICE_OUT_HDMI_EARC,                 // 0x00040001u,
     AUDIO_DEVICE_OUT_SPDIF,                     // 0x00080000u
+    VX_ROCKCHIP_OUT_SPDIF0,
     AUDIO_DEVICE_OUT_FM,                        // 0x00100000u
     AUDIO_DEVICE_OUT_AUX_LINE,                  // 0x00200000u
     AUDIO_DEVICE_OUT_SPEAKER_SAFE,              // 0x00400000u
@@ -247,6 +249,7 @@ static CONST_ARRAY audio_devices_t
AUDIO_DEVICE_IN_ALL_ARRAY[] = {
     AUDIO_DEVICE_IN_BLUETOOTH_SCO_HEADSET,      // 0x80000008u
     AUDIO_DEVICE_IN_WIRED_HEADSET,              // 0x80000010u
     AUDIO_DEVICE_IN_HDMI,                       // 0x80000020u, IN_AUX_DIGITAL
+    VX_ROCKCHIP_IN_HDMI0,
     AUDIO_DEVICE_IN_TELEPHONY_RX,               // 0x80000040u, IN_VOICE_CALL
     AUDIO_DEVICE_IN_BACK_MIC,                   // 0x80000080u
     AUDIO_DEVICE_IN_REMOTE_SUBMIX,              // 0x80000100u
diff --git a/audio/include/system/audio-hal-enums.h b/audio/include/system/audio-
hal-enums.h
index bb00e819..bde599f6 100644
--- a/audio/include/system/audio-hal-enums.h
+++ b/audio/include/system/audio-hal-enums.h
@@ -340,6 +340,7 @@ enum {
     V(AUDIO_DEVICE_OUT_BLUETOOTH_A2DP_HEADPHONES, 0x100u) \
     V(AUDIO_DEVICE_OUT_BLUETOOTH_A2DP_SPEAKER, 0x200u) \
     V(AUDIO_DEVICE_OUT_HDMI, 0x400u) \
+    V(VX_ROCKCHIP_OUT_HDMI0, 0x401u) \
     V(AUDIO_DEVICE_OUT_ANLG_DOCK_HEADSET, 0x800u) \
     V(AUDIO_DEVICE_OUT_DGTL_DOCK_HEADSET, 0x1000u) \
     V(AUDIO_DEVICE_OUT_USB_ACCESSORY, 0x2000u) \
@@ -350,6 +351,7 @@ enum {
     V(AUDIO_DEVICE_OUT_HDMI_ARC, 0x40000u) \
     V(AUDIO_DEVICE_OUT_HDMI_EARC, 0x40001u) \
     V(AUDIO_DEVICE_OUT_SPDIF, 0x80000u) \
+    V(VX_ROCKCHIP_OUT_SPDIF0, 0x80001u) \
     V(AUDIO_DEVICE_OUT_FM, 0x100000u) \
     V(AUDIO_DEVICE_OUT_AUX_LINE, 0x200000u) \
     V(AUDIO_DEVICE_OUT_SPEAKER_SAFE, 0x400000u) \
@@ -369,6 +371,7 @@ enum {
     V(AUDIO_DEVICE_IN_BLUETOOTH_SCO_HEADSET, AUDIO_DEVICE_BIT_IN | 0x8u) \
     V(AUDIO_DEVICE_IN_WIRED_HEADSET, AUDIO_DEVICE_BIT_IN | 0x10u) \
     V(AUDIO_DEVICE_IN_HDMI, AUDIO_DEVICE_BIT_IN | 0x20u) \
+    V(VX_ROCKCHIP_IN_HDMI0, AUDIO_DEVICE_BIT_IN | 0x21u) \
     V(AUDIO_DEVICE_IN_TELEPHONY_RX, AUDIO_DEVICE_BIT_IN | 0x40u) \
     V(AUDIO_DEVICE_IN_BACK_MIC, AUDIO_DEVICE_BIT_IN | 0x80u) \
     V(AUDIO_DEVICE_IN_REMOTE_SUBMIX, AUDIO_DEVICE_BIT_IN | 0x100u) \
```

以上是针对RK3588上hdmi1/spdif1/hdmirx1新增的设备类型，用户可参考如上补丁添加设备类型。设备类型命名规则如下：

```
#输入设备
VX_ROCKCHIP_IN_用户自定义
#输出设备
VX_ROCKCHIP_OUT_用户自定义
```

通过上述方式添加定义后，此设备类型将能够被sdk使用，用户可依据功能需求，在框架/HAL中使用。

1.4 应对GMS/EDLA的修改

通过1.3的方式添加设备后，vts测试可以通过，但是cts-on-gsi相关测试项会fail，原因是gsi当中没有包含新增设备类型，导致解析audio_policy_configuration.xml失败。需要添加如下修改：

system/media

```
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Tue Oct 10 09:58:54 2023 +0800

    MultiAudio: Load audiopolicy xml by priority

    priority: multihal > singlehal > normal

    multihal: audio_policy_configuration_multihal.xml
    singlehal: audio_policy_configuration_singlehal.xml
    normal: audio_policy_configuration.xml

    Fix following fail in cts on gsi:
    APM_AudioPolicyManager: Unable to find audio module for submix, aborting
    mix 0 registration

    Signed-off-by: shine.liu <shine.liu@rock-chips.com>
    Change-Id: Iae7d04f81ddb2c19ba4a665e1653fe028b8377a2

diff --git a/audio/include/system/audio_config.h
b/audio/include/system/audio_config.h
index c4b4e6be..2278bdb0 100644
--- a/audio/include/system/audio_config.h
+++ b/audio/include/system/audio_config.h
@@ -67,6 +67,12 @@ static inline std::string
audio_find_readable_configuration_file(const char* fil
  */
 static inline std::string audio_get_audio_policy_config_file() {
     static constexpr const char *apmXmlConfigFileName =
"audio_policy_configuration.xml";
+//-----------------------rk code----------
+    static constexpr const char *apmMultiAudioXmlConfigFileName =
+            "audio_policy_configuration_multihal.xml";
+    static constexpr const char *apmSingleAudioXmlConfigFileName =
+            "audio_policy_configuration_singlehal.xml";
+//----------------------------------------
     static constexpr const char *apmA2dpOffloadDisabledXmlConfigFileName =
             "audio_policy_configuration_a2dp_offload_disabled.xml";
     static constexpr const char *apmLeOffloadDisabledXmlConfigFileName =
@@ -98,6 +104,19 @@ static inline std::string
audio_get_audio_policy_config_file() {
         audioPolicyXmlConfigFile = audio_find_readable_configuration_file(
                 apmBluetoothLegacyHalXmlConfigFileName);
     }
+//-----------------------rk code----------
```

```
+    std::string audioMultiPolicyXmlConfigFile =
audio_find_readable_configuration_file(
+            apmMultiAudioXmlConfigFileName);
+    std::string audioSinglePolicyXmlConfigFile =
audio_find_readable_configuration_file(
+            apmSingleAudioXmlConfigFileName);
+
+    if (audioPolicyXmlConfigFile.empty()) {
+        if (!audioMultiPolicyXmlConfigFile.empty())
+            return audioMultiPolicyXmlConfigFile;
+        if (!audioSinglePolicyXmlConfigFile.empty())
+            return audioSinglePolicyXmlConfigFile;
+    }
+//-----------------------------------------
     return audioPolicyXmlConfigFile.empty() ?
             audio_find_readable_configuration_file(apmXmlConfigFileName) :
audioPolicyXmlConfigFile;
 }
```

device/rockchip/common

```
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Thu Oct 19 10:25:05 2023 +0800

    MultiAudio: Load audiopolicy xml by priority

    priority: multihal > singlehal > normal

    multihal: audio_policy_configuration_multihal.xml
    singlehal: audio_policy_configuration_singlehal.xml
    normal: audio_policy_configuration.xml

    multihal is used by MultiAudio.
    singlehal is used by General.
    normal is used by GSI.

    Signed-off-by: shine.liu <shine.liu@rock-chips.com>
    Change-Id: Ibeaa6aa5c56d8441c629f76411672185761ea212
```

2. 多声卡同声输出

sdk代码中已经包含同声输出功能相应代码，用户可打开对应宏开关使用对应功能。

2.1 同声输出

用户使用该功能，需打开宏OPEN_ALL_DEVICES_SIMULT

frameworks/av

```
diff --git a/services/audiopolicy/enginedefault/src/Engine.cpp
b/services/audiopolicy/enginedefault/src/Engine.cpp
index bdd6dc1ae8..e3da88616b 100644
--- a/services/audiopolicy/enginedefault/src/Engine.cpp
+++ b/services/audiopolicy/enginedefault/src/Engine.cpp
@@ -25,7 +25,7 @@
 #endif

 // open all the devices simultaneously in one hal
-#define OPEN_ALL_DEVICES_SIMULT 0
+#define OPEN_ALL_DEVICES_SIMULT 1

 #include "Engine.h"
 #include <android-base/macros.h>
```

2.2 speaker音量可调节，其它音量固定

用户使用该功能，需要打开宏FIX_VOL_AND_SEND_VOL_TO_HAL和SET_VOL_IN_HAL

frameworks/av

hardware/rockchip/audio/tinyalsa_hal

```
diff --git a/services/audioflinger/Threads.cpp
b/services/audioflinger/Threads.cpp
index 1b5abe8ca0..65ac674b07 100644
--- a/services/audioflinger/Threads.cpp
+++ b/services/audioflinger/Threads.cpp
@@ -19,7 +19,7 @@
 #define LOG_TAG "AudioFlinger"
 // #define LOG_NDEBUG 0
 #define ATRACE_TAG ATRACE_TAG_AUDIO
-#define FIX_VOL_AND_SEND_VOL_TO_HAL 0
+#define FIX_VOL_AND_SEND_VOL_TO_HAL 1

 #include "Configuration.h"
 #include <math.h>
```

```
diff --git a/tinyalsa_hal/Android.mk b/tinyalsa_hal/Android.mk
index 37600b2..bbf7400 100755
--- a/tinyalsa_hal/Android.mk
+++ b/tinyalsa_hal/Android.mk
@@ -128,7 +128,7 @@ LOCAL_C_INCLUDES += $(commonCIncludes)
 LOCAL_HEADER_LIBRARIES += libhardware_headers
```

```
 LOCAL_CFLAGS := $(commonCFlags)
 LOCAL_CFLAGS += -DPRIMARY_HAL
-#LOCAL_CFLAGS += -DSET_VOL_IN_HAL
+LOCAL_CFLAGS += -DSET_VOL_IN_HAL
 LOCAL_SHARED_LIBRARIES := $(commonSharedLibraries)
 LOCAL_STATIC_LIBRARIES := $(commonStaticLibraries)
 LOCAL_MODULE_TAGS := optional
```

3. 外部通道声音录制问题

如果实际测试发现录屏无法录制外部通道声音，可能是权限问题。

权限问题官方介绍https://developer.android.google.cn/guide/topics/media/platform/av-capture?hl=zh-cn

3.1 用户开发外部通道预览应用

3.1.1 AudioTrack允许被录制

通过setAllowedCapturePolicy(AudioAttributes.ALLOW_CAPTURE_BY_ALL)设置，可参考如下代码:

base/media/java/android/media/AudioStream.java

```
            m_out_trk = new AudioTrack(
                    new AudioAttributes.Builder()
                        .setUsage(AudioAttributes.USAGE_MEDIA)
                        .setContentType(AudioAttributes.CONTENT_TYPE_MUSIC)

.setAllowedCapturePolicy(AudioAttributes.ALLOW_CAPTURE_BY_ALL)
                        .build(),
                        audioFormat, m_out_buf_size, AudioTrack.MODE_STREAM,
AudioManager.AUDIO_SESSION_ID_GENERATE);
```

3.1.2 应用权限

```
dumpsys media.audio_policy
```

通过dump信息可以查看应用是否allowPlaybackCapture，如下所示:

```
Allow playback capture log:
  Package manager errors: 0
  - uid= 1000, allowPlaybackCapture=true , packageName=com.android.keychain
  - uid= 1000, allowPlaybackCapture=true , packageName=com.android.dynsystem
  - uid= 1000, allowPlaybackCapture=true , packageName=com.android.server.telecom
  - uid= 1000, allowPlaybackCapture=false, packageName=com.cghs.stresstest
  - uid= 1000, allowPlaybackCapture=true , packageName=com.android.inputdevices
  - uid= 1000, allowPlaybackCapture=true , packageName=com.android.location.fused
```

```
  - uid= 1000, allowPlaybackCapture=true ,
packageName=com.android.wallpaperbackup
  - uid= 1000, allowPlaybackCapture=false,
packageName=com.android.rockchip.camera2
  - uid= 1000, allowPlaybackCapture=true ,
packageName=com.android.providers.settings
  - uid= 1000, allowPlaybackCapture=false, packageName=com.rockchip.devicetest
  - uid= 1000, allowPlaybackCapture=true ,
packageName=com.example.partnersupportsampletvinput
  - uid= 1000, allowPlaybackCapture=true , packageName=com.android.localtransport
  - uid= 1000, allowPlaybackCapture=false,
packageName=android.rockchip.update.service
  - uid= 1000, allowPlaybackCapture=true , packageName=com.android.settings
  - uid= 1000, allowPlaybackCapture=true , packageName=android
  - uid= 1000, allowPlaybackCapture=false, packageName=com.android.rk
  - uid=10052, allowPlaybackCapture=true , packageName=android.rk.RockVideoPlayer
  - uid=10070, allowPlaybackCapture=true , packageName=com.android.systemui
```

如果应用对应的allowPlaybackCapture=false，playback音频将不允许被录制。

如需启用捕获功能，请在应用的 `manifest.xml` 文件中添加

`android:allowAudioPlaybackCapture="true"`。

注意：应用的音频能否被捕获也取决于应用的 `targetSdkVersion`。 默认情况下，如果应用以 Android 9.0 及之前的版本为目标平台，则不允许捕获播放的音频。如需启用它，请在应用的 `manifest.xml` 文件中添加 `android:allowAudioPlaybackCapture="true"`。 默认情况下，以 Android 10（API 级别 29）或更高版本为目标平台的应用允许捕获其音频。如需停用捕获播放的音频，请在应用的 `manifest.xml` 文件中添加 `android:allowAudioPlaybackCapture="false"`。

3.2 sdk自带的hdmiin.apk

sdk自带的hdmiin.apk属于system应用，一旦system应用有一个allowPlaybackCapture=false，那么外部通道的声音将不能被录制，解决补丁如下：

```
commit e0a87c8fc42dd462afe5971bcb37aba8fbcf4373
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Fri Dec 1 14:36:25 2023 +0800

    AudioPolicy: allowPlaybackCapture for system app.

    Signed-off-by: shine.liu <shine.liu@rock-chips.com>
    Change-Id: Ief820112cbfed7043507ce04d4c7f15bce057d66

diff --git a/services/audiopolicy/service/AudioPolicyInterfaceImpl.cpp
b/services/audiopolicy/service/AudioPolicyInterfaceImpl.cpp
index 2e7b3ffb84..49025b5b1c 100644
--- a/services/audiopolicy/service/AudioPolicyInterfaceImpl.cpp
+++ b/services/audiopolicy/service/AudioPolicyInterfaceImpl.cpp
@@ -355,7 +355,8 @@ Status AudioPolicyService::getOutputForAttr(const
media::AudioAttributesInternal

     if (!mPackageManager.allowPlaybackCapture(VALUE_OR_RETURN_BINDER_STATUS(
         aidl2legacy_int32_t_uid_t(attributionSource.uid)))) {
```

```
-        attr.flags = static_cast<audio_flags_mask_t>(attr.flags |
AUDIO_FLAG_NO_MEDIA_PROJECTION);
+        if (attributionSource.uid != 1000)
+            attr.flags = static_cast<audio_flags_mask_t>(attr.flags |
AUDIO_FLAG_NO_MEDIA_PROJECTION);
     }
     if (((attr.flags &
(AUDIO_FLAG_BYPASS_INTERRUPTION_POLICY|AUDIO_FLAG_BYPASS_MUTE)) != 0)
             && !bypassInterruptionPolicyAllowed(attributionSource)) {
```